

Project Requirement Document

DAY 1

AI + NLP Project

AI Job Description Skill Extraction System

Automatically identify the job role and required skills from a Job Description using NLP.

1. Problem Statement

Recruiters receive hundreds of Job Descriptions (JDs), but manually identifying the job role and required skills is time-consuming and inconsistent. The goal is to build an AI system that automatically extracts structured information from unstructured JD text.

2. Business Objective

Create a smart recruitment solution that helps HR teams:

- Reduce manual screening time
- Standardize skill extraction
- Improve candidate-job matching
- Generate structured recruitment data

3. Technical Objective

Develop an NLP pipeline that:

- Accepts a JD as input
- Detects the job role
- Extracts technical and soft skills
- Returns JSON output
- Can be deployed using Flask or FastAPI

The required technology stack in the task includes Python, Pandas, NumPy, NLTK, spaCy, Scikit-learn, Transformers, Matplotlib, Seaborn, Power BI/Tableau, and Flask/FastAPI.

4. Understanding the NLP Concepts

Concept | Meaning

Keyword Extraction | Finds important words like Python, SQL

Entity Extraction | Identifies named entities such as skills, tools, roles

Classification | Predicts the job category (Data Scientist, Java Developer)

Information Extraction | Extracts structured fields like role, skills, experience

These are the core concepts listed in the Day 1 task.

5. Project Input

The input is a plain-text Job Description.

Example

We require a Data Scientist with Python, Pandas, NumPy, Scikit-learn, SQL and AWS experience.

6. Expected Output

JSON

```
{
  "role": "Data Scientist",
  "skills": [
    "Python",
    "Pandas",
    "NumPy",
    "Scikit-learn",
    "SQL",
    "AWS"
  ]
}
```

This output format matches the example in the PDF.

7. Technologies

Python - NLP development

Jupyter Notebook - Experimentation

Pandas & NumPy - Data processing

NLTK & spaCy - Text preprocessing

Transformers - Advanced NLP

Power BI / Tableau - Analytics dashboard

8. Success Criteria

The project is successful if it can:

- Correctly identify the job role
- Extract 90%+ of required skills from a JD
- Produce structured JSON output
- Process JDs in real time through an API

9. Future Scope (Resume-worthy)

You can improve this into a production-level AI project by adding:

- 1. Skill recommendations for missing skills**
- 2. Experience extraction (3+ years, 5 years)**
- 3. Education extraction**
- 4. Salary & location extraction**
- 5. Resume vs JD matching score**
- 6. AI suggestions to improve the Job Description**

Deliverable

File name: Project_Requirement_Document.pdf

This is a strong Day 1 document that aligns with the internship task and is suitable for submission.